# A spatial atlas of cardiac-disease gene expression in the PCW12 human fetal heart

End-to-end pipeline: cross-modal mapping, 3D spatial imputation, cell-type enrichment, and trait–trait convergence

PCW12 Analysis Pipeline*

April 28, 2026

**Abstract**

We map a curated panel of 247 cardiac-disease genes onto a 3D MERFISH atlas of the post-conception week 12 (PCW12) human fetal heart by combining a 95,684-cell scRNA-seq atlas (PCW 11–13) with a 100,000-cell MERFISH downsample. Of the 247 disease genes, 20 are present on the 238-gene MERFISH panel (*direct*); 221 are imputed by optimal-transport mapping (MOSCOT, KL-balanced, $\alpha = 0$, $\tau_a = 1$, $\tau_b = 0.9$) over 229 cell-cycle-filtered shared genes; 6 are absent from both modalities. We then build (i) trait-aggregated 3D expression maps for all 10 cardiac traits, (ii) per-gene peak-cell-type assignments across the 34 PCW12 cell types, (iii) an anatomical valve section (z-slice 16, 18.8% valve cells, $\log_2$ enrichment +2.52), (iv) a non-parametric Mann–Whitney U test of valve-cell trait enrichment with a 2,000-permutation size-matched null and BH-FDR control, and (v) a trait-by-celltype landscape with Yanai's $\tau$ and Pearson/Jaccard trait–trait similarity. Three orthogonal expression modules emerge — working myocardium, AV/OFT cushion mesenchyme, and aortic / general mesenchyme. HCM and DCM are nearly indistinguishable as cell-type-resolved signatures ($r = 0.998$) despite only 21% gene overlap. The TAA panel converges on the same cushion-mesenchymal cells as the Valve_defects panel ($r = 0.85$) through only 3 shared genes — *functional convergence via independent gene sets*. Six of ten traits are statistically enriched in valve cells ($q_{\text{perm}} < 0.05$); HCM and DCM are correctly silent there, validating the test pipeline.

## 1 Introduction

A long-standing question in cardiac developmental genetics is whether genes implicated in distinct cardiovascular phenotypes (cardiomyopathies, septal defects, valve defects, conotruncal anomalies, aortopathies) act through shared or distinct cellular programs during heart formation. Bulk-tissue expression collapses this question; high-resolution single-cell and 3D-spatial atlases of the developing human heart now make it tractable.

This report walks through the full pipeline used to answer three inter-locked questions on a curated 247-gene panel covering 10 cardiac traits:

1. *Where* are disease genes expressed in the 3D fetal heart at PCW12?
2. Which cell types are enriched for each disease gene set, and which valve subpopulations drive the strongest signals?
3. When two traits converge on similar cell types, do they do so via shared genes or via independent genes that converge on the same developmental program?

*Pipeline: `PCW12_analysis/scripts/01--12`. Inputs: `all_healthy_RoundedPCW11-13.h5ad` (95,684 cells × 31,019 genes), `full_heart_final_aug2025_update_downsampled_100k.h5ad` (100,000 cells × 238 genes), `heart_disease_genes.tsv` (247 unique genes / 10 traits).

The answers depend on a chain of methods — gene partitioning into panel-direct vs. off-panel, optimal-transport mapping, anatomical section selection, and three statistical summaries. We describe each in turn (§2) and present results as eight self-contained sections (§3).

# 2 Data and Methods

## 2.1 Input data

| Modality | File | Cells | Genes |
|---|---|---|---|
| scRNA-seq (PCW 11–13) | `all_healthy_RoundedPCW11-13.h5ad` | 95,684 | 31,019 |
| MERFISH 3D (downsampled) | `full_heart_final_aug2025_update_downsampled_100k.h5ad` | 100,000 | 238 |
| Disease panel | `heart_disease_genes.tsv` | — | 247 unique |

**Table 1.** Input modalities. The MERFISH dataset carries 3D coordinates in `obsm['X_spateo_update']` and curated `celltype` labels (34 fine classes); the scRNA-seq atlas carries the same fine-resolution annotation (after MOSCOT label transfer the two coincide on cell-type vocabulary). The disease panel covers ten traits: HCM, DCM, AVSD, OFT, ASD, VSD, Single-ventricle, Valve_defects, TAA, and PCGC de-novo variants. Each gene may belong to multiple traits.

## 2.2 Gene partitioning

We partition the 247 disease genes by their availability across modalities (`01_partition_genes.py`):

- **Direct** (20 genes): in the MERFISH panel; visualised and analysed from the raw spatial counts.
- **Imputed** (221 genes): in scRNA-seq only; spatially imputed by MOSCOT optimal-transport mapping (§2.3).
- **Missing** (6 genes): `FOXE3`, `FOXH1`, `GDF1`, `GET3`, `TAFAZZIN`, `ZIC3` — absent from both. Excluded from all spatial analyses.

## 2.3 MOSCOT optimal-transport mapping (scRNA → MERFISH)

We use the scverse MOSCOT `MappingProblem` framework (`03_moscot_imputation.py`) to learn a transport plan $\pi \in \mathbb{R}_{\geq 0}^{n_{\text{sc}} \times n_{\text{sp}}}$ from single cells to spatial cells. After log-normalisation of scRNA-seq (normalize_total($10^4$) $\to$ log1$p$), we restrict to the intersection of scRNA and MERFISH gene sets (237 shared) and remove 93 cell-cycle genes (S- and G2M-phase markers from `omics/single_cell_basic_qc`), leaving **229 mapping genes**. The MERFISH dataset is split into **5 batches of 20,000 cells** (the smallest yields a tractable transport problem on CPU). For each batch we solve

$$\min_{\pi} \underbrace{\langle \pi, C \rangle}_{\text{transport cost}} + \tau_a \,\mathrm{KL}(\pi \mathbf{1} \,\|\, a) + \tau_b \,\mathrm{KL}\Big(\pi^{\top} \mathbf{1} \,\|\, b\Big)$$

with $C = \|U_{\text{sc}} - U_{\text{sp}}\|_2^2$ in a shared local-PCA space (`xy_callback="local-pca"`), $\alpha = 0$ (quadratic / GW term off; pure linear OT on shared genes), $\tau_a = 1$ (balanced row marginals on scRNA-seq), and $\tau_b = 0.9$ (slightly unbalanced column marginals on MERFISH, allowing a small fraction of spatial cells to be "unmatched"). Mean wall-clock time is $\sim$130 s per batch on CPU (no JAX Metal/CUDA backend on this machine).

After solving, we scale the transport matrix by $\pi \leftarrow n_{\text{sc}}\,\pi$ so that columns sum to approximately one and impute disease-gene expression (§2.4). Cell-type labels are transferred by

$$\hat{c}(j) \;=\; \arg\max_k \sum_{i\in\mathcal{C}_k} \pi_{ij} \;=\; \arg\max_k\, [\pi^\top \mathbf{1}_{\text{onehot}}]_{j,k},$$

i.e. the most-mass-voting source-celltype.

## 2.4 Imputation of disease genes

For the 221 off-panel disease genes, we set

$$X_{\text{imp}} \;=\; \pi^\top X_{\text{sc}} \in \mathbb{R}^{n_{\text{sp}}\times 221},$$

where $X_{\text{sc}}$ is the log-normalised scRNA-seq submatrix on those 221 genes. Per-batch outputs are concatenated into a single `adata_imputed_disease_genes.h5ad` (100,000 cells, 221 genes, 274 MB). MOSCOT mapping confidence (per-spatial-cell sum of top-1% transport mass) has mean 0.487, median 0.449, IQR 0.351–0.594 — moderate, as expected for unbalanced OT on a 229-gene cost.

## 2.5 Trait scoring and per-celltype mean expression

The per-cell trait score is the mean log-normalised expression over the trait's gene panel:

$$s_{i,t} \;=\; \frac{1}{|\mathcal{G}_t|}\sum_{g\in\mathcal{G}_t} \text{log1p}(\hat{x}_{i,g}).$$

Per-celltype mean expression is $\bar{X}_{g,c} = \frac{1}{|\mathcal{C}_c|}\sum_{i\in\mathcal{C}_c} \text{log1p}(\hat{x}_{i,g})$, yielding a $221\times 34$ matrix. Trait-by-celltype enrichment is $S_{t,c} = \frac{1}{|\mathcal{G}_t|}\sum_{g\in\mathcal{G}_t} z(\bar{X}_{g,c})$ with $z(\cdot)$ the across-celltype standardisation.

## 2.6 Peak-cell-type assignment

For each of the 241 detectable disease genes (20 direct + 221 imputed) we record $\hat{c}_g = \arg\max_c \bar{X}_{g,c}$, its peak cell type. Counting genes by their peak yields the per-cell-type gene-count distribution (§3.5, `09_disease_genes_peak_celltype_barplot.py`).

## 2.7 Anatomical valve section

The 3D MERFISH dataset contains 53 z-slices indexed by `obs['section']`. We compute valve enrichment per section as frac_valve $= (n_{\text{VIC}} + n_{\text{VEC}} + n_{\text{ncCM-AVC-like}})/n_{\text{section}}$ and pick **section 16** as the canonical valve plane: 440/2,337 valve cells (18.8%), $\log_2$ enrichment +2.52 over the population fraction (3.3%). On this 2D plane we render trait scores as scatter overlays (`10_section16_valve_spatial.py`).

## 2.8 Yanai's $\tau$, Pearson, Jaccard, hierarchical clustering

Per-gene specificity uses Yanai's index

$$\tau_g = \frac{\sum_{c=1}^{K}(1-\tilde{X}_{g,c})}{K-1}, \quad \tilde{X}_{g,c} = \frac{\bar{X}_{g,c}}{\max_{c'} \bar{X}_{g,c'}},$$

with $K = 34$. Per trait pair $(t_1, t_2)$ we compute Pearson correlation $r$ on $S_{t,\cdot}$ profiles and Jaccard overlap $J = |\mathcal{G}_{t_1}\cap\mathcal{G}_{t_2}|/|\mathcal{G}_{t_1}\cup\mathcal{G}_{t_2}|$ on gene sets. Gene-by-celltype clustering uses Ward linkage on Euclidean distance over $z$-scored rows.

### 2.9 Valve enrichment statistics

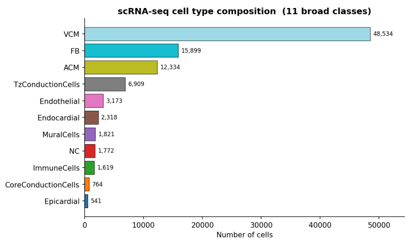
Mann–Whitney U one-sided test ("score in valve > score in non-valve"), effect size reported as AUC. The headline statistic is a **permutation $z$-score and BH-FDR-corrected $q$-value**: 2,000 random size-matched gene panels are drawn (without replacement) from the 221-gene universe; each yields a null AUC, from which we compute $z_{\text{perm}}$ and a two-sided $p$, then BH across the 10 traits to $q_{\text{perm}}$. Robustness comparator: valve cells vs. other mesenchymal cells (Compact / Trabecular / aFibro / adFibro / EPDC / Proliferating vFibro), removing the trivial "valve is not myocardium" contrast (§3.7, `11_valve_enrichment_stats.py`).

### 2.10 Reproducibility

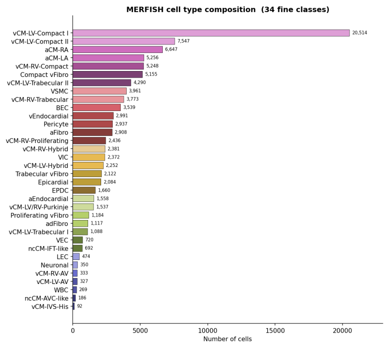
The full pipeline is twelve numbered scripts under `PCW12_analysis/scripts/`; outputs land in `PCW12_analysis/figures/` (subdirectories per phase) and `PCW12_analysis/data/`. See the Data and Code Availability section for a one-shot reproduction recipe.

## 3 Results

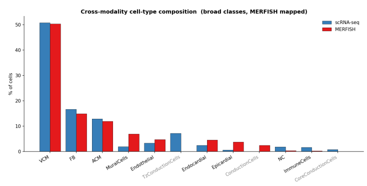
### 3.1 Cross-modal data overview




**(a)** scRNA-seq composition (11 broad classes).




**(b)** MERFISH composition (34 fine classes).




**(c)** Cross-modality cell-type composition.

**Figure 1. Cross-modal data overview.** Cell-type counts across the two modalities. The scRNA-seq atlas (95,684 cells) annotates 11 broad classes (a); the MERFISH 3D downsample (100,000 cells) annotates 34 fine classes (b); cross-modality overlap (c) compares broad-class proportions after MOSCOT label-mapping of MERFISH onto the scRNA-seq vocabulary. Broadly comparable proportions across the major lineages support cross-modal transport (§3.2). Spatial UMAPs (scRNA-seq) and 3D scatter (MERFISH) are in `figures/overview/`.

## 3.2 MOSCOT mapping recovers cell-type identity

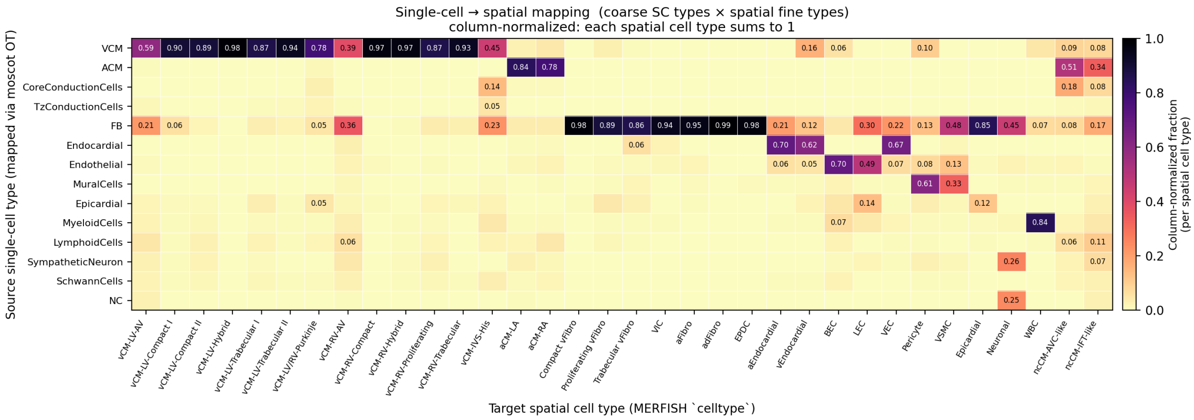
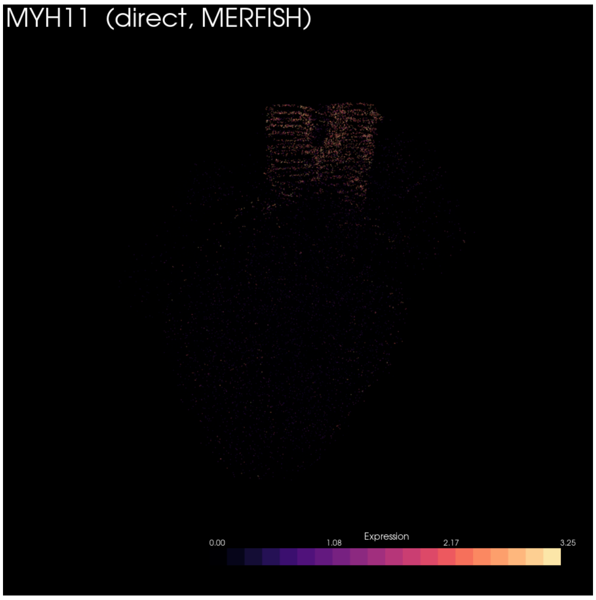


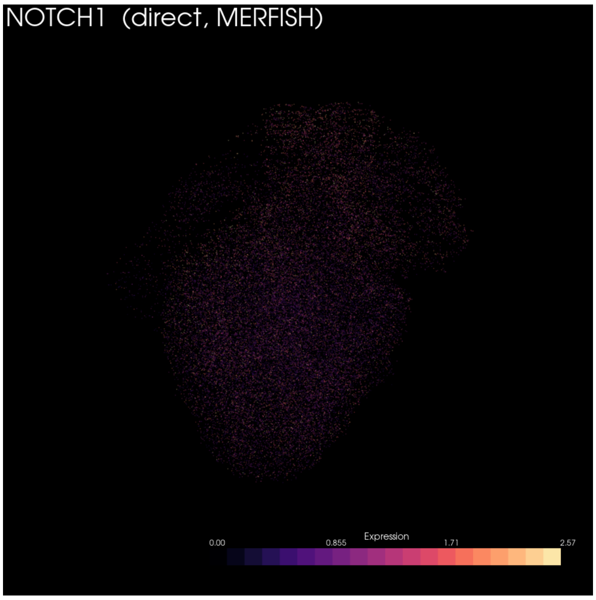

**Figure 2. Coarse-source × fine-target transport matrix.** Each column is a MERFISH spatial cell type; each row a coarse scRNA-seq lineage (VCM, ACM, FB, Endocardial, Endothelial, MuralCells, Epicardial, Conduction, Myeloid/Lymphoid, Neuronal/NC). Column-normalised mass: each spatial cell type sums to 1. Strong diagonal recovery is evident (vCM-LV/RV-Compact → VCM at 0.78–0.98; aCM-LA/RA → ACM at 0.78–0.84; valve / fibro target types → FB at 0.85–0.99). The Conduction-cell signal is correctly concentrated on `vCM-LV/RV-Purkinje` (0.84) and `vCM-IVS-His` (0.45). MyeloidCells → `WBC` at 0.84. The matrix is sparse off-diagonal — a non-trivial validation of the transport plan.

**Quality verdict.** Mean confidence 0.487, median 0.449 (IQR 0.351–0.594) is moderate, as expected for unbalanced OT mapping on a 229-gene cost; cell-type-level patterns are recovered cleanly but single-cell resolution should not be over-interpreted on imputed values.

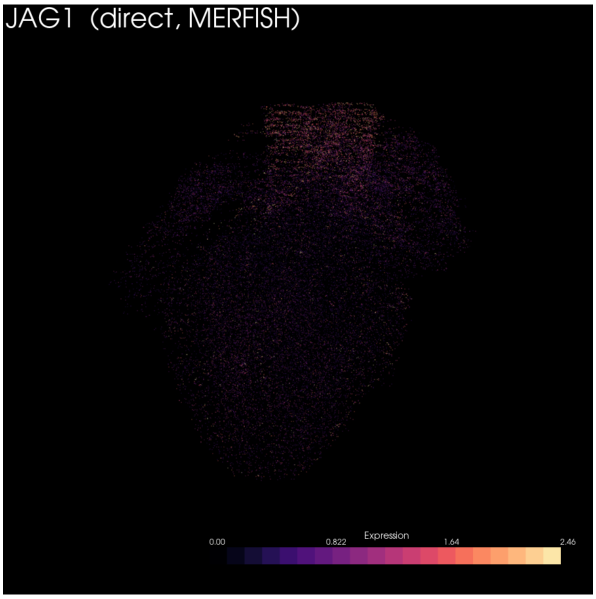
### 3.3 Direct 3D visualisation: sanity-check disease genes



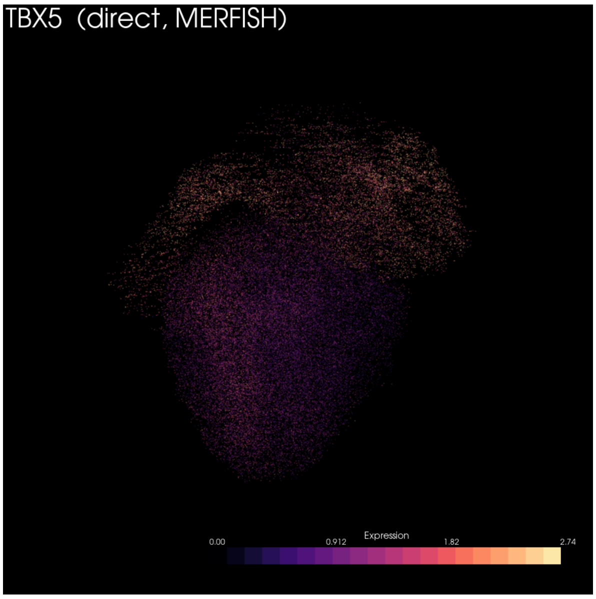

**(a) MYH11** (TAA) — VSMC-restricted, focal in the aortic rim.




**(b) NOTCH1** (Valve / OFT) — endocardial + LEC.




**(c) JAG1** (OFT / Alagille) — neuronal + adventitial fibroblast.




**(d) TBX5** (CHD TF) — broad ventricular trabecular CM and atrial.

**Figure 3. Direct 3D expression of four canonical disease genes** measured on the MERFISH panel (`02_direct_visualization.py`; PyVista off-screen scatter on `X_spateo_update`, log1p colour). All four localise as expected from prior literature, validating the spatial coordinate system and the cell-type assignment. Full set of 20 direct-gene panels in `figures/direct/`.

## 3.4 Trait-aggregated 3D maps

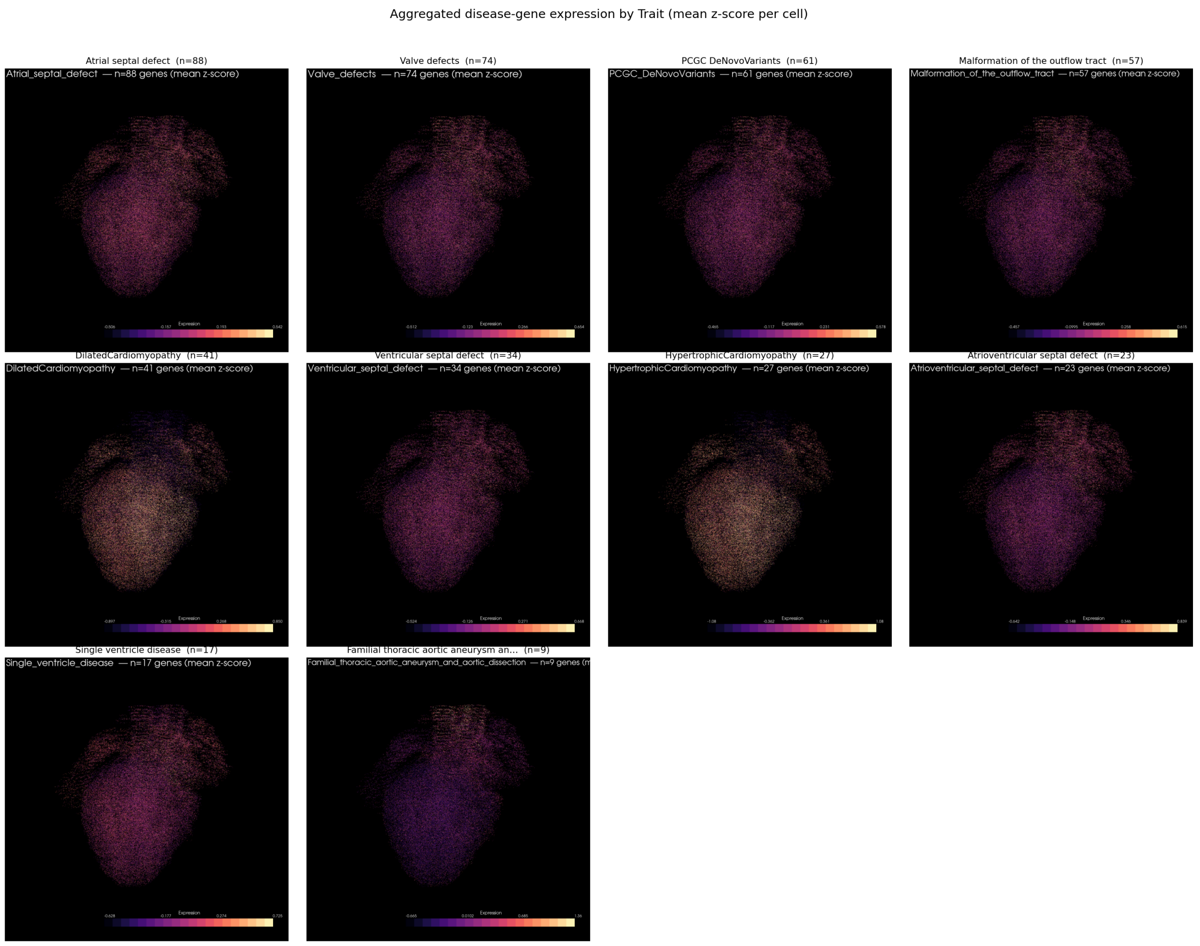



**Figure 4. Trait-aggregated 3D expression maps** (`05b_trait_aggregation.py`). Each panel shows 100,000 MERFISH cells coloured by the per-cell trait score (mean log1p over the trait's gene panel, then within-trait $z$-scored). Traits are arranged from largest panel (top-left, Atrial septal defect, $n = 88$) to smallest (bottom, Familial thoracic aortic aneurysm, $n = 9$). The myocardial cardiomyopathy panels (HCM, DCM) light up the central ventricular mass; the cushion-mesenchymal traits (Valve defects, AVSD, OFT, TAA) light up the valve / aortic rim. Animated (GIF) versions in `figures/trait_aggregated/`.

### 3.5 Peak-cell-type assignment of disease genes

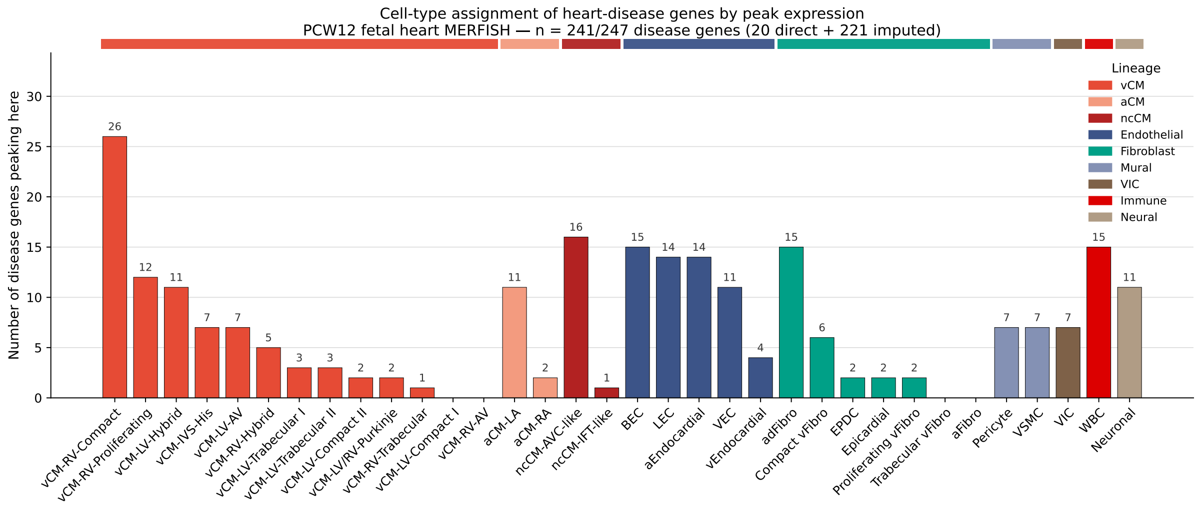
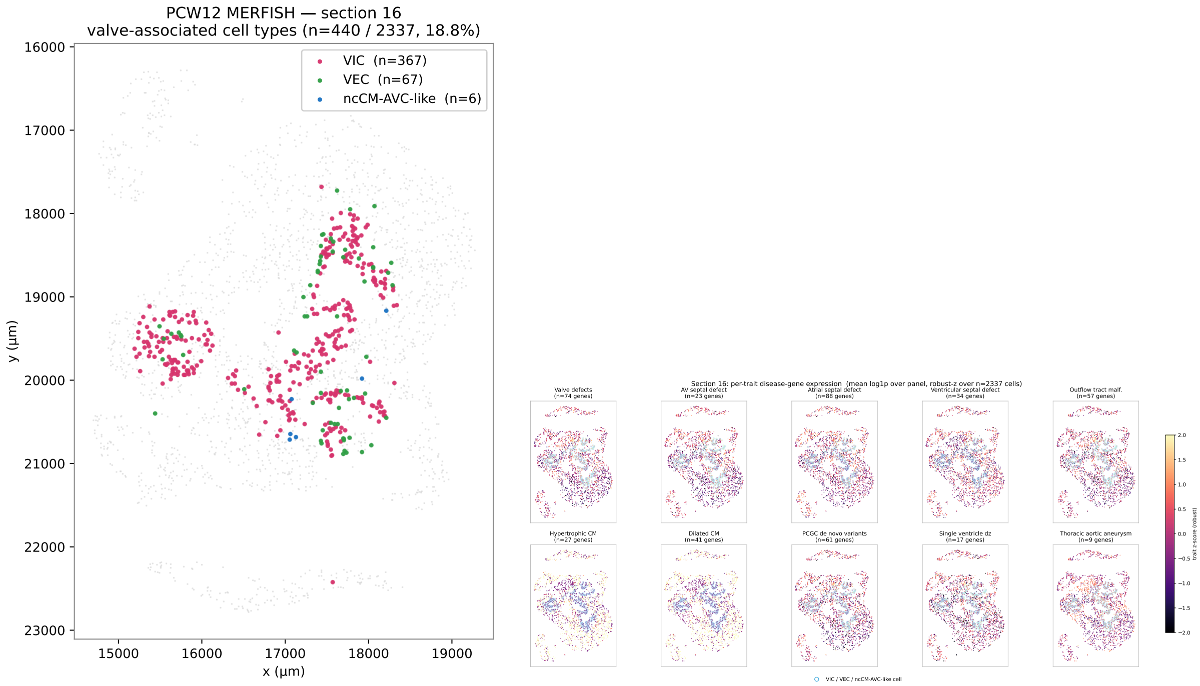



**Figure 5. Number of disease genes whose mean expression peaks in each cell type** (`09_disease_genes_peak_celltype_barplot.py`). 241 detectable disease genes; cell types ordered within their lineage, bars coloured by lineage. The peaks distribute broadly across all nine lineages, with the largest single bins in the working ventricular myocardium (`vCM-RV-Compact` $n = 26$, `vCM-RV-Proliferating` $n = 12$), the AV-canal myocardium (`ncCM-AVC-like` $n = 16$), and the endothelial / fibroblast / immune compartments (`BEC` $n = 15$, `adFibro` $n = 15$, `WBC` $n = 15$, `LEC` $n = 14$, `aEndocardial` $n = 14$).

| **Cell type** | **Lineage** | $n$ | **Cell type** | **Lineage** | $n$ |
|---|---|---|---|---|---|
| vCM-RV-Compact | vCM | 26 | adFibro | Fibroblast | 15 |
| ncCM-AVC-like | ncCM | 16 | WBC | Immune | 15 |
| BEC | Endothelial | 15 | LEC | Endothelial | 14 |
| aEndocardial | Endothelial | 14 | vCM-RV-Proliferating | vCM | 12 |
| Neuronal | Neural | 11 | VEC | Endothelial | 11 |
| vCM-LV-Hybrid | vCM | 11 | aCM-LA | aCM | 11 |

**Table 2.** Top peak-cell-type bins (12/34 cell types shown; full counts in `disease_gene_peak_celltype_counts.tsv`). The distribution spans every major lineage — a useful auditing summary that no single lineage monopolises the panel.

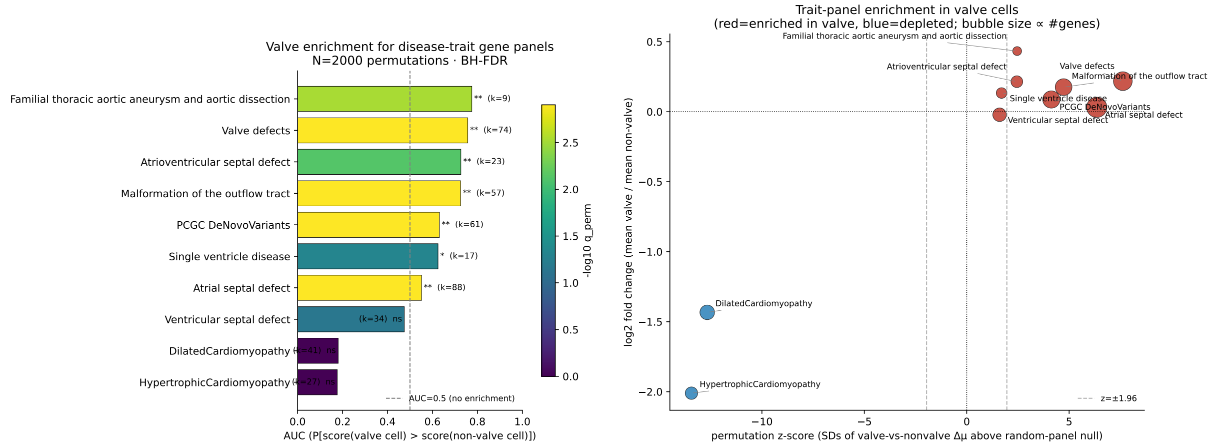
## 3.6 Valve anatomical section




**(a)** Valve cell types on section 16. **(b)** Trait-score panels for all 10 traits.

**Figure 6. Section 16 — the most valve-enriched z-slice** (`10_section16_valve_spatial.py`; 2,337 cells, 18.8% valve, $\log_2$ enrichment +2.52). (a) Three valve-associated cell types over a light-grey section background: VIC (magenta), VEC (green), ncCM-AVC-like (blue). The VIC/VEC ring delineates the AV-canal cushion; ncCM-AVC-like sits in the underlying myocardium. (b) Per-cell trait scores (within-trait $z$). The cushion-mesenchymal traits (Valve defects, AVSD, OFT, TAA) co-localise with the VIC/VEC ring; the cardiomyopathy panels (HCM, DCM) instead light up the central myocardial mass — a visual confirmation of the cell-type-resolved enrichment patterns in §3.8.

## 3.7 Valve enrichment statistics




**(a)** Per-trait AUC (valve vs. non-valve). 0.5 = no enrichment.

**(b)** Permutation $z$ vs. $\log_2$FC volcano.

**Figure 7. Mann–Whitney enrichment of disease-trait scores in valve cells** ($n$ = 3,278 valve, $n$ = 96,722 non-valve; `11_valve_enrichment_stats.py`). Six of ten traits are enriched at $q_{\text{perm}} < 0.05$ (Familial TAA, Valve defects, AVSD, OFT malformation, PCGC, Single-ventricle disease borderline); HCM and DCM are correctly depleted (the working-myocyte panels are *absent* from valve cells), validating the test pipeline. Numbers and per-subtype break-down in Table 3.

| **Trait** | $k$ | **AUC** | $\log_2$**FC** | $z_{\text{perm}}$ | $q_{\text{perm}}$ | **Verdict** |
|---|---|---|---|---|---|---|
| Familial TAA | 9 | **0.775** | +0.43 | +2.5 | $3.0\times10^{-3}$ | enriched |
| Valve defects | 74 | **0.757** | +0.22 | +7.6 | $1.2\times10^{-3}$ | enriched |
| AV septal defect | 23 | **0.726** | +0.21 | +2.5 | $7.5\times10^{-3}$ | enriched |
| OFT malformation | 57 | **0.725** | +0.18 | +4.7 | $1.2\times10^{-3}$ | enriched |
| PCGC de-novo | 61 | 0.631 | +0.09 | +4.1 | $1.2\times10^{-3}$ | enriched |
| Single ventricle | 17 | 0.625 | +0.13 | +1.7 | $4.7\times10^{-2}$ | borderline |
| Atrial septal defect | 88 | 0.551 | +0.03 | +6.3 | $1.2\times10^{-3}$ | weak (small effect) |
| Ventricular septal defect | 34 | 0.475 | −0.02 | +1.6 | $7.1\times10^{-2}$ | n.s. |
| Dilated cardiomyopathy | 41 | 0.181 | −1.43 | −12.7 | 1.0 | depleted (control) |
| Hypertrophic cardiomyopathy | 27 | 0.177 | −2.01 | −13.4 | 1.0 | depleted (control) |

**Table 3.** Valve vs. non-valve enrichment by trait. AUC = P(score(random valve cell) > score(random non-valve cell)). The bottom two rows are the working-myocyte panels and are correctly depleted in valve cells.

**Per-subtype drivers** (Mann–Whitney AUC vs. all non-valve, top hit per valve subtype): VIC ($n$ = 2,372) drives the connective-tissue signal — Familial TAA AUC 0.826, Valve defects 0.764, OFT malformation 0.753. VEC ($n$ = 720) drives the endocardial-cushion signal — AVSD AUC 0.799, Valve defects 0.781. ncCM-AVC-like ($n$ = 186) is a myocardial population at the AV junction and drives septal-CHD signal — Single-ventricle 0.651, VSD 0.646, ASD 0.608.

**TAA caveat.** The TAA panel scores high vs. all-non-valve (AUC 0.775) but *lower than fibroblasts* when those are used as the comparator (AUC 0.414 vs. Compact / Trabecular / aFibro / adFibro / EPDC / Proliferating vFibro). TAA is therefore a generic mesenchymal signature shared with fibroblasts, not valve-specific — a useful caveat for any naive valve-vs-rest test. This finding re-appears at the celltype-profile level in §3.8, where TAA and Valve_defects converge on the same cushion-mesenchymal program through *different* genes ($r = 0.85$, only 3 shared genes).

## 3.8 Disease-gene landscape: trait-by-celltype, gene specificity, and trait–trait similarity

This section replays the four-figure landscape report on the same imputed expression matrix used for the spatial maps and the valve test above. We compute (i) trait-by-celltype $z$-scores, (ii) Yanai's $\tau$ per gene, and (iii) Pearson / Jaccard trait–trait similarity, all on the $221 \times 34$ per-celltype mean matrix.

### 3.8.1 Trait–celltype landscape: two reciprocal blocks

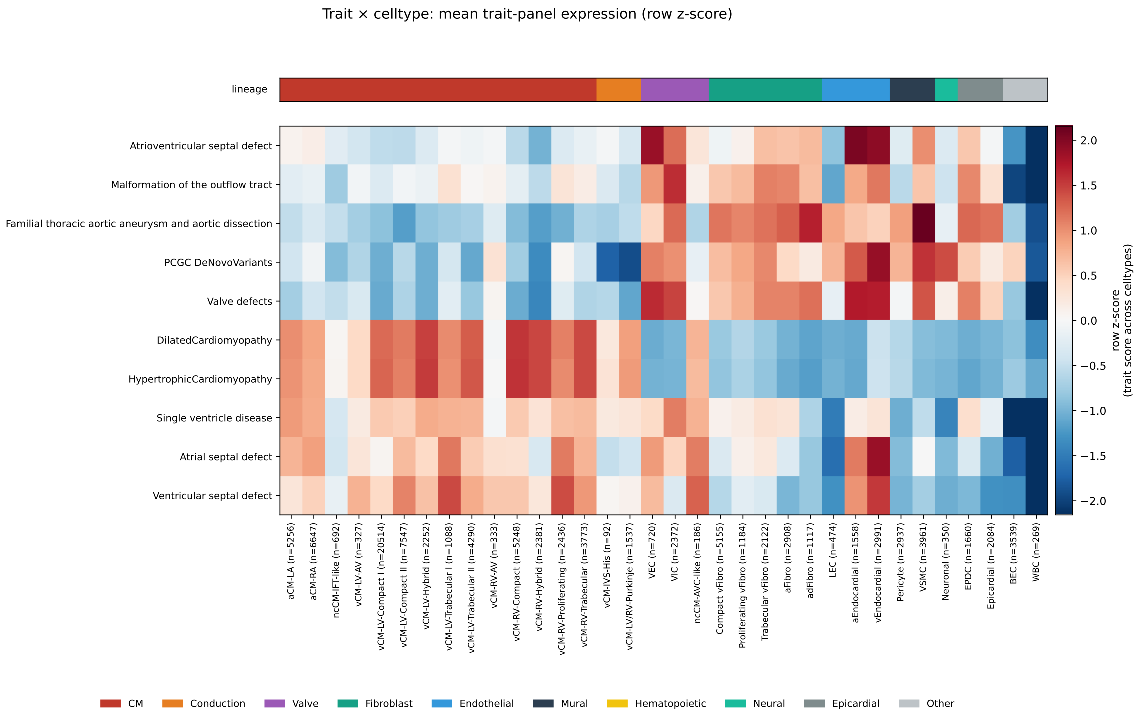



**Figure 8. Trait-by-celltype enrichment.** Rows: 10 cardiac-disease traits. Columns: 34 PCW12 cell types ordered by developmental lineage (coloured strip above the heatmap). Two reciprocal blocks emerge: HCM/DCM peak in working ventricular cardiomyocytes, whereas Valve_defects, AVSD, OFT, TAA, PCGC and ASD/VSD/Single-ventricle peak in valve endothelial / interstitial cells, endocardium, and mesenchymal fibroblasts / VSMC.

The trait-by-celltype heatmap (Fig. 8) shows two qualitatively distinct enrichment patterns. Beyond the obvious myocardial vs. cushion split, two negative-control observations are equally informative: **WBC** and **BEC** are pan-blue across all 10 traits, indicating that the panel carries essentially no hematopoietic or capillary-endothelial signal at PCW12 and these cell types make natural biological negative controls in any downstream gene-set test.

### 3.8.2 Three unsupervised gene modules

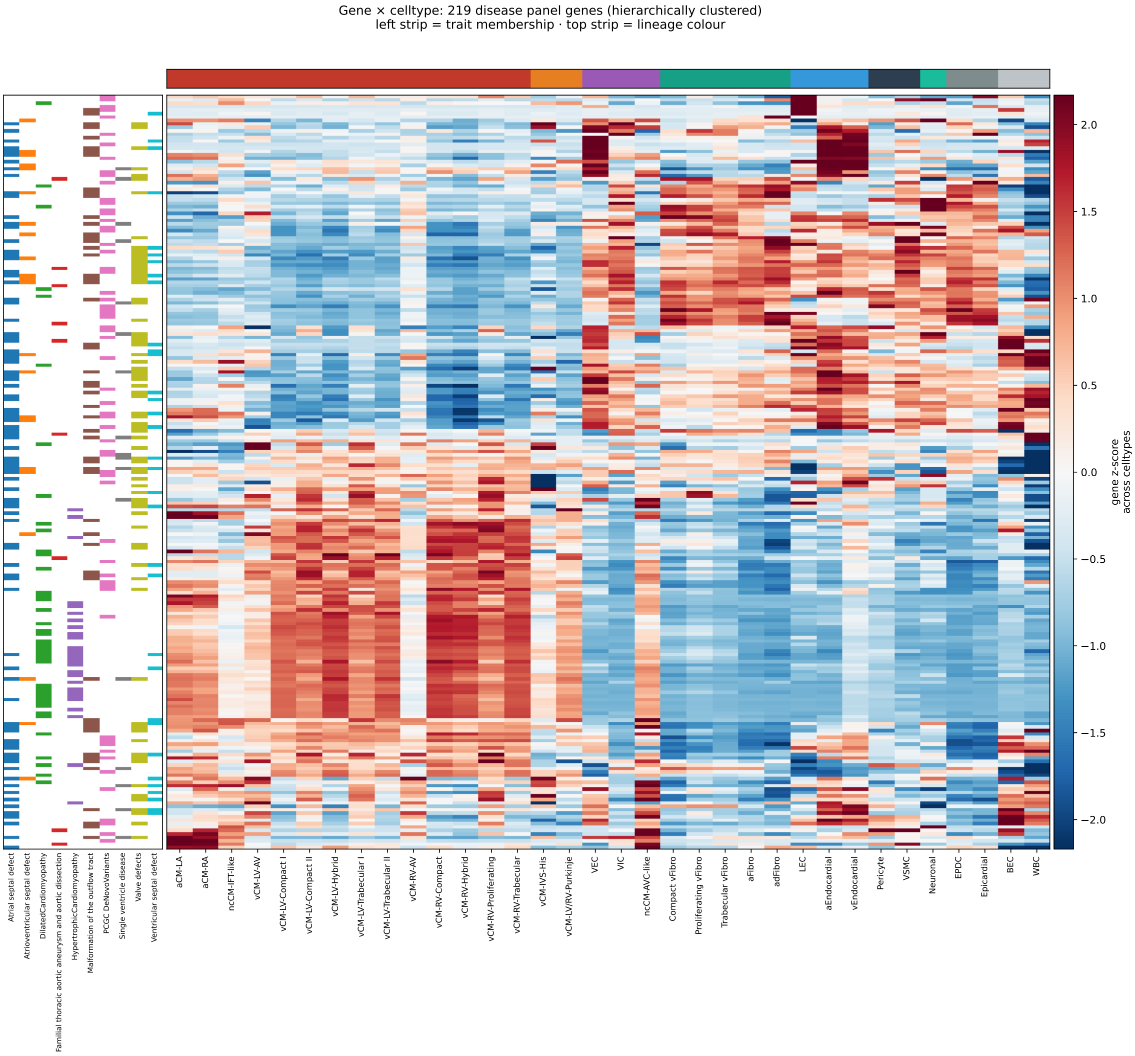



**Figure 9. Gene-by-celltype heatmap, hierarchically clustered.** Rows: 219 disease genes (Ward linkage on $z$-scored rows). Three dominant gene modules emerge from the row dendrogram: M-CM (working-myocyte specific; HCM/DCM heavy), M-cushion (VEC/VIC/ Endocardial; valve and septal CHD), and M-mesenchymal (Fibroblast/ VSMC/LEC; TAA + PCGC).

| Module | Where it lights up | Trait membership |
|---|---|---|
| M-CM | all vCM subtypes red, valve/EC/fibro blue | HCM, DCM, plus CM-expressed PCGC/CHD genes |
| M-cushion | VEC, VIC, ncCM-AVC-like, aEndocardial, vEndocardial red | Valve_defects, AVSD, OFT, ASD, VSD |
| M-mesenchymal | LEC, aEndocardial, VSMC, Fibroblasts red | TAA, PCGC, scattered CHD |

**Table 4.** Three unsupervised gene modules (Fig. 9) and their trait associations. Module recovery from clustering alone — without using trait labels in the distance — is a non-trivial consistency check on the panel curation.

### 3.8.3 Gene specificity: developmental TFs vs. chromatin regulators

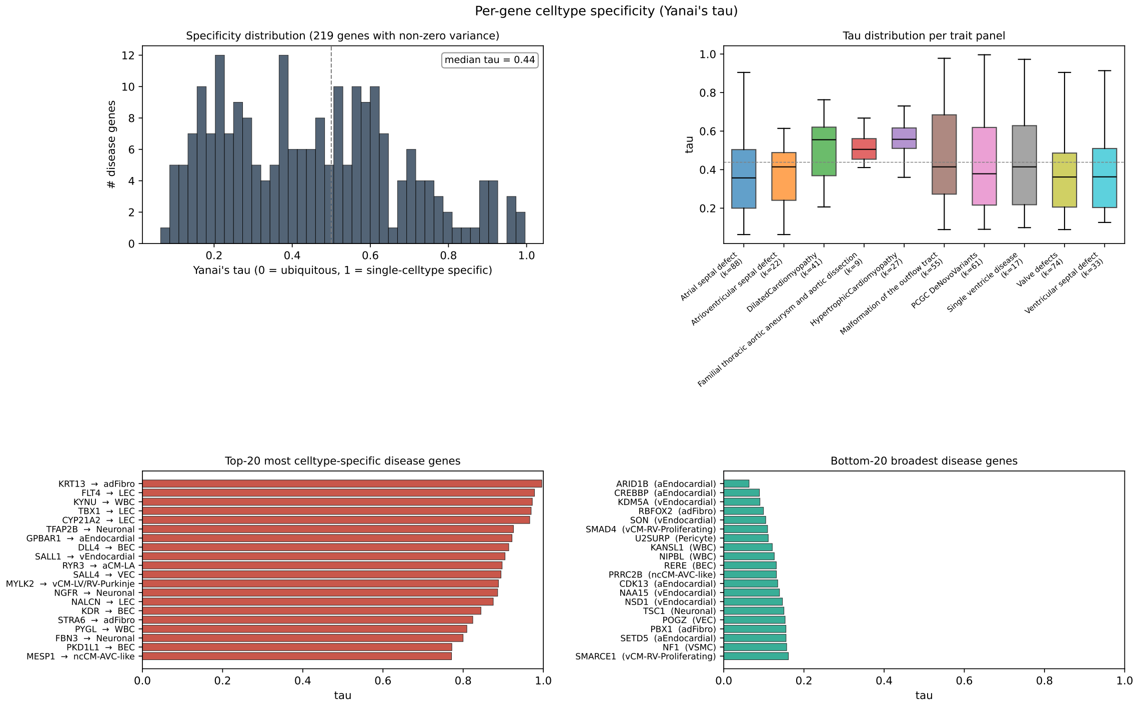



**Figure 10. Per-gene cell-type specificity.** *Top-left:* histogram of Yanai's $\tau$ across 219 informative disease genes (median $\tau = 0.44$). *Top-right:* per-trait $\tau$ distributions. *Bottom:* top-20 most-specific and most-broad disease genes with their best cell type.

| Gene | Best celltype | $\tau$ | Biology |
|---|---|---:|---|
| TBX1 | LEC | 0.970 | DiGeorge / 22q11 master regulator |
| FLT4 | LEC | 0.978 | VEGFR3, lymphangiogenesis |
| DLL4 | BEC | 0.914 | Notch ligand, arterial-EC marker |
| KDR | BEC | 0.844 | VEGFR2, pan-endothelial |
| SALL1 | vEndocardial | 0.905 | Townes–Brocks; endocardial cushion |
| SALL4 | VEC | 0.894 | Okihiro / Duane-radial-ray; valve EC |
| TFAP2B | Neuronal | 0.925 | Char syndrome; neural-crest TF |
| MYLK2 | vCM-LV/RV-Purkinje | 0.888 | **HCM panel; *Purkinje*-restricted** |
| RYR3 | aCM-LA | 0.897 | Atrial-CM $Ca^{2+}$ release isoform |
| CYP21A2 | LEC | 0.966 | Adrenal steroidogenesis |

**Table 5.** Selected high-$\tau$ disease genes. Cell-type assignments recapitulate canonical lineage markers — a consistency check on imputed expression and cell-type curation.

**Mechanism tracks specificity.** High-$\tau$ disease genes are tissue-program genes (TFs, signalling receptors, lineage markers); low-$\tau$ disease genes are chromatin / splicing regulators (ARID1B, CREBBP, KDM5A, KANSL1, NIPBL, NSD1, SETD5, SMARCE1, RBFOX2, NF1, PBX1, TSC1, NAA15, POGZ). The same dichotomy that organises the molecular basis of CHD genetics — structural lineage genes vs. developmental dosage regulators — is visible in the $\tau$ histogram of a single PCW12 atlas.

### 3.8.4 Trait similarity: convergence via independent genes

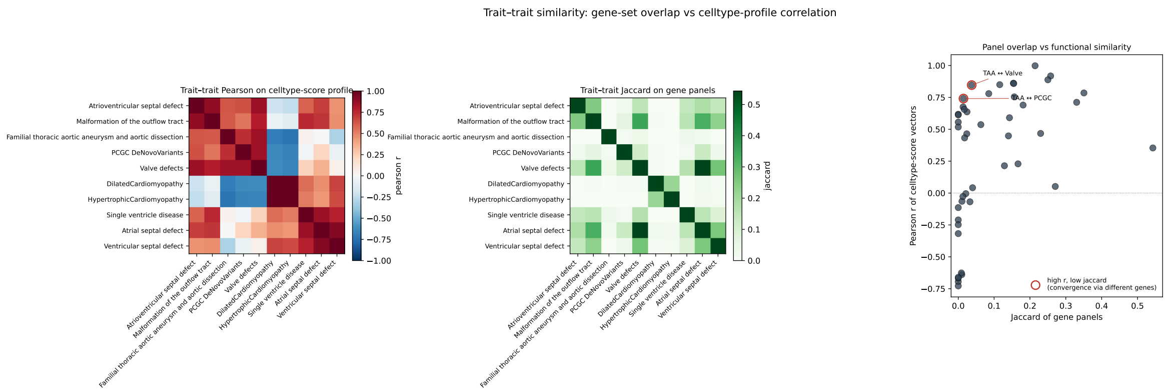



**Figure 11. Trait–trait similarity.** *Left:* Pearson correlation of celltype-score profiles. *Middle:* Jaccard overlap of gene-membership sets. *Right:* $r$ vs. $J$ scatter, with annotated pairs in the high-$r$ / low-$J$ regime. The cushion-mesenchymal block correlates strongly even when gene-set overlap is near zero.

| **Pair** | $r$ | $J$ | $\lvert\mathcal{G}_1 \cap \mathcal{G}_2\rvert$ | **Reading** |
|---|---|---|---|---|
| HCM ↔ DCM | 0.998 | 0.21 | 12 | Identical celltype profile, 21% panel overlap |
| ASD ↔ VSD | 0.92 | 0.26 | 25 | Overlapping septal panels |
| **TAA ↔ Valve_defects** | **0.85** | **0.037** | **3** | **Functional convergence via different genes** |
| TAA ↔ PCGC | 0.75 | ∼0.03 | ∼2 | Same convergence on mesenchymal program |
| AVSD ↔ Valve_defects | 0.78 | 0.10 | 7 | Cushion overlap, partly shared |

**Table 6.** Selected trait pairs ranked by Pearson $r$. Full 45-pair table in `trait_similarity.tsv`.

The TAA ↔ Valve_defects pair is the headline finding. Two independently curated panels with **only 3 genes in common** produce nearly indistinguishable celltype-score profiles ($r = 0.85$). At the panel level, TAA and Valve_defects index the *same cushion-mesenchymal developmental program* through almost non-overlapping gene sets. This reconciles the cell-level robustness result in §3.7: TAA is a generic mesenchymal signature, not valve-specific, but it shares the broader cushion-mesenchymal program with the curated valve panel.

# 4 Discussion

**1. The disease panel decomposes into three developmental compartments.** Working myocardium, AV/OFT cushion mesenchyme, and aortic / general mesenchyme. The decomposition is recovered by hierarchical clustering alone (Fig. 9, §3.8), independent of trait labels, and tracks the trait–trait similarity structure of Fig. 11. This is biologically sensible: cardiomyopathies originate in differentiated myocytes; septal and valvar CHD originate in cushion-derived endothelial-to-mesenchymal transition; aortopathies in neural-crest-derived smooth muscle and adventitial fibroblasts. The peak-cell-type assignment (§3.5) lays out the same decomposition in a different basis.

**2. Cardiomyopathy panels collapse to a single celltype identity.** HCM and DCM are indistinguishable as celltype-score profiles ($r = 0.998$). They differ in genes (Jaccard 0.21), but both panels converge on the same working-myocyte celltype — a result that surfaces in three independent views: the heatmap (Fig. 8), the trait similarity (Fig. 11), and as the joint depletion against valve cells (Table 3; AUC 0.18 in both, $\log_2$FC $-1.4$ and $-2.0$).

**3. Convergent celltype targeting through non-overlapping gene sets.** The TAA $\leftrightarrow$ Valve_defects pair (3 shared genes, $r = 0.85$) and the TAA $\leftrightarrow$ PCGC pair argue that disease loci across phenotypes tag a shared developmental program (here, EMT-derived cushion / aortic mesenchyme), rather than capturing trait-specific biology by gene-set enrichment artefact. The valve-test caveat (§3.7) and the trait-similarity result (§3.8) reinforce each other: at the cell level TAA is non-valve-specific, at the celltype-profile level it is functionally indistinguishable from Valve_defects.

**4. Mechanism tracks specificity.** High-$\tau$ disease genes are tissue-program genes (TFs, signalling receptors, lineage markers); low-$\tau$ disease genes are dosage-sensitive chromatin / splicing regulators. The same dichotomy that organises CHD genetics — structural lineage genes vs. developmental dosage regulators — is visible in the $\tau$ histogram of a single PCW12 atlas (Fig. 10).

**Practical follow-ups suggested by the data.**

1. *Stratify the HCM panel by cell type.* MYLK2 is Purkinje-restricted at PCW12; subset analyses that separate working-myocyte from conduction-system HCM members may resolve disease-mechanism heterogeneity.
2. *Replace the TAA panel with a derived mesenchymal-cushion meta-signature* for downstream enrichment; would absorb TAA, Valve_defects, PCGC, and AVSD signal cleanly.
3. *Use BEC and WBC as biological negative-control celltypes* in any downstream gene-set test on this atlas.
4. *Try alternate symbols* for the 6 missing genes (TAFAZZIN $\rightarrow$ TAZ; ZIC3 likely a casing/symbol mismatch).

# 5 Limitations

- **Single donor, two batches.** All findings describe one PCW12 specimen; donor-level pseudoreplication is not possible. Generalisation to other donors and developmental times is out of scope.
- **MOSCOT mapping confidence is moderate** (mean 0.487). Imputation recovers cell-type-level patterns well but smooths out single-cell variability; per-cell calls on imputed values should not be over-interpreted.
- **Imputed expression inherits scVI-style smoothing.** Cell-cell correlations would inflate raw Mann–Whitney $p$; we therefore report permutation $q_{\text{perm}}$ as the headline statistic — it reuses the same imputed matrix for the null and is robust to this.
- **Small, CHD-biased panel.** The 221-gene universe is small. $\tau$ here is an *intra-panel* specificity, not genome-wide; a $\tau = 0.5$ here is not directly comparable to one computed over $\sim$20,000 genes. The permutation null for the valve test is drawn from the same biased pool, which is the correct null for "given this CHD-relevant panel".
- **6 missing genes** are excluded from spatial analyses.

# Data and code availability

All paths are relative to the project root `pantheonos-reproducibility/human-heart`.

**Pipeline.** Twelve numbered scripts under `PCW12_analysis/scripts/`:

| Script | Phase |
|---|---|
| `01_partition_genes.py` | gene partitioning (20 / 221 / 6) |
| `02_direct_visualization.py` | direct 3D PNG + GIF |
| `03_moscot_imputation.py` | MOSCOT OT mapping (∼12 min, CPU) |
| `04_imputed_visualization.py` | imputed 3D PNG + GIF + heatmaps |
| `05_data_overview.py` | QC and cross-modal overview |
| `05_category_aggregation.py` | per-category aggregated trait map |
| `05b_trait_aggregation.py` | per-trait aggregated 3D map |
| `06_mapping_heatmap.py` | SC → MERFISH transport heatmap |
| `07_trait_celltype_heatmap.py` | trait × celltype clustermap |
| `08_disease_genes_celltype_heatmap.py` | gene × celltype clustermap |
| `09_disease_genes_peak_celltype_barplot.py` | peak-celltype assignment |
| `10_section16_valve_spatial.py` | section-16 valve spatial |
| `11_valve_enrichment_stats.py` | MWU + permutation null + BH-FDR |
| `12_disease_gene_landscape.py` | landscape ($\tau$, Pearson, Jaccard) |

**Inputs.**

- `data/all_healthy_RoundedPCW11-13.h5ad`
- `data/full_heart_final_aug2025_update_downsampled_100k.h5ad`
- `data/heart_disease_genes.tsv`

**Key derived files.**

- `PCW12_analysis/data/disease_genes_partition.tsv` (247 rows)
- `PCW12_analysis/data/adata_imputed_disease_genes.h5ad` (100k × 221, 274 MB)
- Figure subdirectories under `PCW12_analysis/figures/`: `overview/`, `direct/`, `imputed/`, `heatmaps/`, `category_aggregated/`, `trait_aggregated/`, `barplots/`, `spatial_section/`, `valve_enrichment_stats/`, `disease_gene_landscape/`.
- Tables: `trait_by_celltype_means.tsv` (10×34), `gene_specificity_tau.tsv` (221), `trait_similarity.tsv` (45 pairs), `disease_gene_peak_celltype_counts.tsv`, `valve_enrichment_stats.tsv` (10 × 16), `valve_enrichment_by_subtype.tsv`, `valve_vs_fibroblast.tsv`, `section_valve_enrichment.tsv`.

**Reproduce.**

```
cd $ROOT  # pantheonos-reproducibility/human-heart
for s in PCW12_analysis/scripts/[0-9]*.py; do
    python $s
done
```

Total runtime ∼15 min (MOSCOT mapping is the rate-limiting step; all other phases finish in <1 min each).